

The chemical synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 21 TO 23 OF WEEK 4

20 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential reaches the end of an axon and stops there. The next cell is across a gap of extracellular fluid, and an electrical signal cannot cross that gap. In this lecture we follow how the signal is turned into a chemical, carried across, and turned back into an electrical signal on the other side. The place where this happens is a chemical synapse. We take the events in strict order, because each step causes the next one, and once you know the order you can predict what happens when any single step fails.

Here is the problem we solve by the end. A five month old girl has been constipated for several days. Now she feeds poorly, her cry is weak, her eyelids droop, and she cannot hold up her head. Within two days she needs a ventilator to breathe. She is alert and she feels every touch. A week earlier her pacifier had been dipped in honey.

Testing narrows it down. She is awake, follows faces and grimaces when her foot is pinched, so her sensory neurons and her brain are working. Action potentials travel down her motor axons at normal speed, so conduction is working. But when a motor nerve is stimulated, the electrical response of the muscle is far smaller than normal. In laboratory studies of muscle poisoned in this way, acetylcholine applied straight onto the muscle still makes it contract normally, so the muscle and its receptors work. Her gut, which is also driven by neurons that release acetylcholine, is hardly moving.

Put those results together. The axon conducts, and the muscle responds when acetylcholine reaches it. So the failure lies between the action potential arriving at the end of the axon and acetylcholine arriving at the muscle's receptors. That stretch is the synapse, and this lecture divides it into nine steps. As you read, decide which step has failed in this baby.

See it move: run one synapse yourself

SLIDE 15 OF THE DECK

This is a working model of the synapse in this lecture. Press Send one action potential and follow the step that is named under the picture. The table on the right reports what happened in numbers, and the graph shows the response of the postsynaptic cell. Then block one step at a time. Each block matches a toxin or drug described later in these notes, so keep the model open and come back to it as you read. The note under the model says which parts are real and which are illustrative.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html)

THREE THINGS TO TRY

- 1 Send one action potential with nothing blocked and read the table. Then turn on Botulinum toxin and send another. Calcium still enters, but vesicles released falls from 3 to 0 and the response falls from 1.00 to 0. Now turn that off and block the Ca^{2+} channels. The postsynaptic cell is just as silent, but this time calcium entry reads no. The readouts upstream of the block are what tell you where the sequence stopped.
- 2 Clear the blocks and block the removal enzyme. The peak of the response rises from 1.00 to about 1.80 and it lasts about 39 ms where it lasted about 28 ms before. Block reuptake as well, so that only diffusion is left, and it rises to about 3.69 and lasts about 75 ms. Compare the solid line on the graph with the dashed line each time.
- 3 Clear the blocks, send a steady train, and move the slider from 10 to 100 action potentials per second. The average response over the train rises from about 0.14 to about 1.35, because more action potentials per second release more transmitter per second. Then block the removal enzyme at 100 per second and watch the response stop returning to baseline.

The parts of a chemical synapse

SLIDE 3 OF THE DECK

A synapse is the place where a neuron passes its signal to another cell. The cell sending the signal is the presynaptic cell, and the cell receiving it is the postsynaptic cell, which may be another neuron, a muscle cell or a gland cell. A chemical synapse has three parts.

The first part is the axon terminal of the presynaptic neuron. This is the output zone from the lecture on neurons. It holds synaptic vesicles, which are small membrane sacs filled with neurotransmitter. Some vesicles are docked against the

membrane that faces the next cell, ready to release, and others wait in reserve farther back. The terminal has mitochondria to supply ATP, and its membrane contains voltage gated Ca^{2+} channels, which are closed at rest.

The second part is the synaptic cleft, a gap of extracellular fluid about 20 to 40 nm wide between neurons, and a little wider at the neuromuscular junction. The two membranes do not touch, which is why the action potential cannot continue across.

The third part is the postsynaptic membrane, which holds receptors. A receptor is a protein that binds one particular transmitter. Astrocytes wrap many synapses in the central nervous system, and they take up some transmitters from the cleft, as you saw in the lecture on glia.

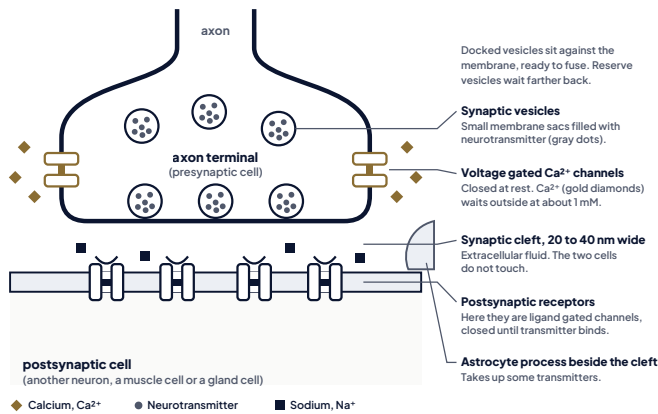


Figure 1. The presynaptic terminal, the synaptic cleft and the postsynaptic membrane. The cleft is drawn much wider than it really is.

Synapse

The place where a neuron passes its signal to another cell.

Presynaptic and postsynaptic

The presynaptic cell sends the signal. The postsynaptic cell receives it.

Synaptic vesicle

A small membrane sac in the axon terminal, filled with neurotransmitter.

Synaptic cleft

The gap of extracellular fluid between the two cells, about 20 to 40 nm wide.

Receptor

A protein in the postsynaptic membrane that binds one particular transmitter and starts the response.

Steps 1 to 5: inside the axon terminal

SLIDES 4 TO 5 OF THE DECK

Step 1. An action potential travels down the axon and depolarizes the membrane of the terminal. Step 2. That depolarization opens voltage gated Ca^{2+} channels. These channels respond to voltage in the same way as the voltage gated sodium channels of the axon, but the ion they pass is calcium.

Step 3. Ca^{2+} enters the terminal. Free calcium is about 1 mM outside the cell and about 0.0001 mM inside, which is a 10,000 fold difference, and the inside of the cell is negative. Both the concentration gradient and the voltage drive calcium in. Worked problem 1 puts numbers on this.

Step 4. The Ca^{2+} that enters binds a sensor protein called synaptotagmin on vesicles that are already docked at the membrane. Step 5. That triggers the SNARE proteins. One SNARE protein sits on the vesicle, and its partners sit on the terminal membrane. They wind around each other and pull the two membranes together until they fuse. The inside of the vesicle is now open to the cleft, and the transmitter empties out. Release by fusion of a vesicle with the cell membrane is exocytosis, which was covered in Week 2. Afterward the vesicle membrane is taken back in by endocytosis and refilled.

Calcium is the link between the electrical signal and the chemical one. Without Ca^{2+} entry there is no release, however many action potentials arrive. It also explains how the frequency code from the last lecture is passed on: more action potentials per second let in more Ca^{2+} , more vesicles fuse, and more transmitter is released.

One classic experiment shows that calcium entry is required. Put the terminal in a bath with no calcium and send an action potential. Steps 1 and 2 happen normally, because the channels respond to voltage and not to calcium. Step 3 fails, because there is no Ca^{2+} outside to enter, and nothing after it happens.

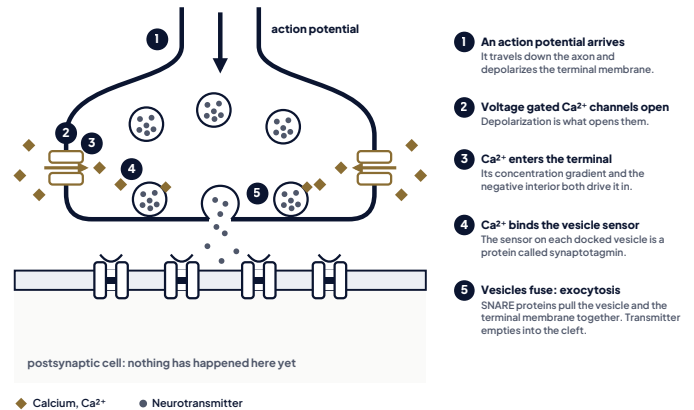


Figure 2. The first five steps. Each one causes the next.

WORKED PROBLEM 1

Free Ca^{2+} is about 1 mM outside the terminal and about 0.0001 mM inside. Find E_{Ca} . Then find the driving force on Ca^{2+} at rest, -70 mV, and at the peak of an action potential, +30 mV, and say which way Ca^{2+} is driven each time.

Use the Nernst equation from the resting potential lecture.
Driving force = membrane potential – equilibrium potential of the ion. For a positive ion, a negative answer means the ion is driven into the cell.

THE WORK

- 1 Write the ratio, outside over inside. $1 \div 0.0001 = 10,000$. The log of 10,000 is 4.
- 2 Calcium carries two positive charges, so $z = +2$. $E_{\text{Ca}} = (61.5 \div 2) \times 4 = 30.75 \times 4 = +123 \text{ mV}$. This is the same value as on the voltage scale in the resting potential lecture.
- 3 The driving force on an ion is the membrane potential minus that ion's equilibrium potential. At rest: $-70 - (+123) = -193 \text{ mV}$. For a positive ion, a negative driving force means the ion is driven **into** the cell.
- 4 At the peak of the action potential: $+30 - (+123) = -93 \text{ mV}$. That is smaller, but it still points inward. The membrane never gets anywhere near +123 mV, so at every moment of an action potential calcium is driven inward.
- 5 So why does calcium not enter at rest, when the driving force is largest? Because its channels are closed. The gradient is always there. The action potential supplies the one thing that was missing, which is an open channel.
- 6 Because there is so little free calcium inside to begin with, a small amount entering raises the concentration right beside the channels a hundredfold or more within a fraction of a millisecond. A signal that starts from almost nothing and rises that fast is easy for the cell to detect, and the vesicle sensor sits right beside those channels.

Voltage gated Ca^{2+} channel

A channel in the terminal membrane that opens when the membrane depolarizes and lets calcium in.

Synaptotagmin

The calcium sensor on a docked vesicle. When calcium binds it, fusion is triggered.

SNARE proteins

Proteins on the vesicle and on the terminal membrane that pull the two membranes together so that they fuse.

Exocytosis

Release of a vesicle's contents by fusion of the vesicle with the cell membrane.

Driving force

Membrane potential minus an ion's equilibrium potential. It tells you how hard, and which way, the ion is driven when a channel opens.

Steps 6 to 9, the delay, and the direction

SLIDES 6 TO 7 OF THE DECK

Step 6. The transmitter diffuses across the cleft. Nothing carries it. Because the gap is only 20 to 40 nm, diffusion takes about a microsecond. Step 7. The transmitter binds receptors on the postsynaptic membrane. Each receptor fits one transmitter, the way a lock fits one key, and the binding is reversible: a transmitter molecule binds, stays for a moment, and comes off again.

Step 8. The postsynaptic cell responds. In the simplest case the receptor is a ligand gated channel, which was covered in the resting potential lecture. With transmitter bound, it opens, Na^+ enters, and the membrane depolarizes a little. That response is a graded potential. Its size depends on how much transmitter was released, it fades with distance, and it has no threshold of its own. Whether it leads to an action potential in the postsynaptic cell is the subject of the next lecture.

Step 9. The transmitter is removed from the cleft. As the concentration in the cleft falls, the receptors empty, the channels close, and the signal ends. The three routes of removal have their own section below.

All of this takes time. The postsynaptic response begins about 0.5 to 1 ms after the action potential reaches the terminal. This is the synaptic delay. Most people guess that the delay is the time taken to diffuse across the cleft, but that is only about a microsecond, less than one percent of it. Most of the delay is spent in the terminal: the voltage gated Ca^{2+} channels take time to open, and then Ca^{2+} has to trigger fusion.

A chemical synapse also works in one direction only. Only the presynaptic side has vesicles ready to release, and only the postsynaptic side has the receptors facing them.

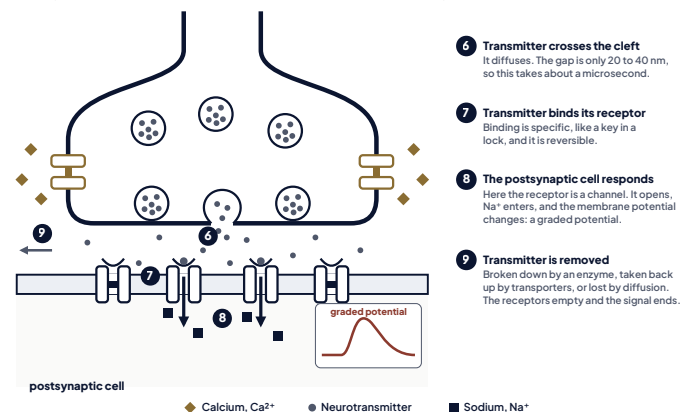


Figure 3. The last four steps. The response in the postsynaptic cell is a graded potential.

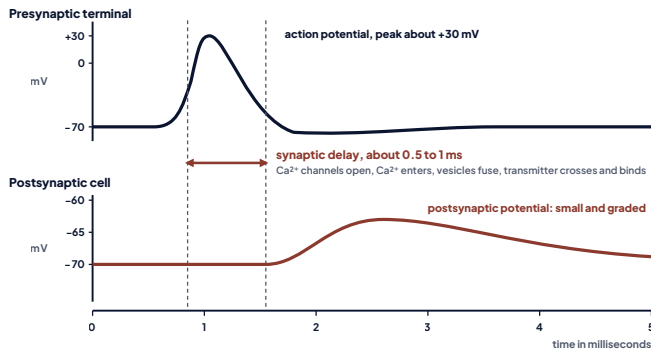


Figure 4. The action potential in the terminal and the response in the postsynaptic cell, on one time axis. Voltages are typical teaching values.

WORKED PROBLEM 2

A pathway has 1 m of axon in total, conducting at 50 m/s. How long does the signal take if the pathway contains 1 synapse? How long if it contains 5?

Use 1 ms for each synaptic delay.

THE WORK

- 1 Conduction time is distance divided by speed. $1 \text{ m} \div 50 \text{ m/s} = 0.02 \text{ s} = \mathbf{20 \text{ ms}}$.
- 2 One synapse adds about 1 ms. Total: $20 + 1 = \mathbf{21 \text{ ms}}$.
- 3 Five synapses add about 5 ms. Total: $20 + 5 = \mathbf{25 \text{ ms}}$. The same length of axon now takes about one fifth longer.
- 4 Put the delay in terms of distance. In 1 ms this axon conducts $50 \text{ m/s} \times 0.001 \text{ s} = 0.05 \text{ m}$. Each synapse costs as much time as **5 cm** of axon, although the cleft itself is only 20 to 40 nm across.
- 5 This is why the fastest reflexes use the fewest synapses. The knee jerk reflex has a single synapse in the spinal cord between the sensory neuron and the motor neuron.
- 6 The delay is a cost of using a chemical synapse. What the cell gets in return is a place where the signal can be made larger or smaller, which is the subject of the next lecture.

Graded potential

A change in membrane potential whose size depends on the strength of the stimulus, here the amount of transmitter. It fades with distance.

Synaptic delay

The time between the action potential reaching the terminal and the start of the postsynaptic response, about 0.5 to 1 ms.

and is over in milliseconds. The nicotinic acetylcholine receptor, the AMPA receptor for glutamate and the GABA-A receptor are examples.

A metabotropic receptor is not a channel. With transmitter bound, it activates a G protein on the inner face of the membrane, which is why it is also called a G protein coupled receptor. The G protein may act on a nearby ion channel directly, or it may switch on an enzyme that makes a second messenger such as cAMP. Second messengers can open channels, close channels, or change other proteins in the cell. More steps make the response slower to start, and it can last from hundreds of milliseconds to minutes. One receptor activates many G proteins, so the signal is also amplified. The muscarinic acetylcholine receptor, every receptor for norepinephrine and dopamine, and most receptors for serotonin are metabotropic.

Now the most important idea in this section. The receptor, and not the transmitter, decides the response. Acetylcholine shows this clearly. At skeletal muscle it binds the nicotinic receptor, whose channel passes both Na^+ and K^+ . A muscle fiber rests near -90 mV , so the driving force on Na^+ is $-90 - 61 = -151 \text{ mV}$, inward, and the driving force on K^+ is $-90 - (-95) = +5 \text{ mV}$, outward. Sodium is driven about 30 times harder, so much more Na^+ enters than K^+ leaves, and the muscle fiber depolarizes and contracts.

At the pacemaker cells of the heart, the same acetylcholine binds the muscarinic receptor. Its G protein opens K^+ channels, K^+ leaves, the cell moves away from threshold, and the pacemaker fires less often, so the heart rate slows. That is one of the actions of the muscarinic receptor in the heart and the easiest one to draw. We return to the others in the cardiac weeks.

So the transmitter is only the message that something has arrived. What the postsynaptic cell does about it depends on which receptor it has put in its membrane and on which ions move as a result. If the result brings the cell toward threshold, the synapse is excitatory. If it holds the cell away from threshold, the synapse is inhibitory. It is like a whistle at a sports field: the same whistle tells one team to start and the other to stop, and the meaning is in the listener.

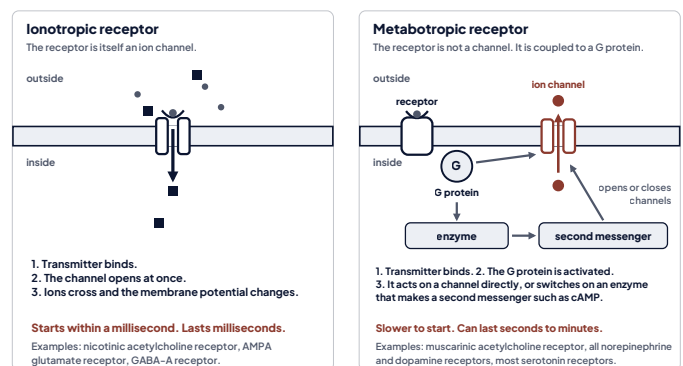


Figure 5. An ionotropic receptor is a channel. A metabotropic receptor works through a G protein.

Two families of receptor, and why the receptor decides

SLIDES 8 TO 9 OF THE DECK

Receptors for neurotransmitters come in two families. An ionotropic receptor is a ligand gated ion channel: the receptor and the channel are the same protein. Transmitter binds, the pore opens, and ions cross. The response begins within a millisecond

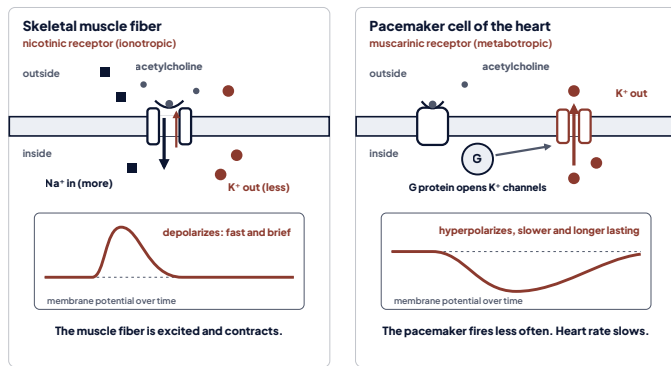


Figure 6. Acetylcholine arriving at two different cells.

Ionotropic receptor

A receptor that is itself an ion channel. Fast and brief.

Metabotropic receptor

A receptor coupled to a G protein. It may act through a second messenger. Slower to start and longer lasting.

Second messenger

A small molecule made inside the cell, such as cAMP, that carries the signal from the receptor to its targets.

Nicotinic and muscarinic receptors

The two families of acetylcholine receptor. Nicotinic receptors are ionotropic. Muscarinic receptors are metabotropic.

Seven classes of neurotransmitter

SLIDES 10 TO 12 OF THE DECK

Silverthorn groups neurotransmitters into seven classes by chemical structure. Acetylcholine is a class of its own. The amines are each made from a single amino acid: dopamine, norepinephrine and epinephrine from tyrosine (these three are the catecholamines), serotonin from tryptophan, and histamine from histidine. The amino acids do most of the fast signaling in the central nervous system: glutamate is the main excitatory transmitter, GABA is the main inhibitory transmitter of the brain, and glycine is the main inhibitory transmitter of the spinal cord. The purines are ATP and adenosine.

Peptides are short chains of amino acids. Substance P excites pain pathways in the spinal cord, and the opioid peptides, the endorphins and enkephalins, inhibit them. Morphine acts on the same receptors. A neuron often releases a peptide together with a small transmitter. The last two classes are unusual. Gases such as nitric oxide, and lipids such as the endocannabinoids, are made at the moment they are needed. They are not stored in vesicles, and they leave the cell by diffusing through its membrane.

The table gives the usual effect of each transmitter and one real place where it acts. Usual effect means the effect at its most common receptors, and the receptor has the final say. The GABA_A receptor and the glycine receptor are Cl⁻ channels. Opening them holds the membrane away from threshold, and the next lecture explains how. A few more facts are worth knowing. Epinephrine is mainly a hormone from the adrenal medulla and a transmitter in only a few brain pathways. Histamine neurons in the hypothalamus promote wakefulness, which is why older antihistamines cause drowsiness. Adenosine is inhibitory in the brain, and caffeine blocks its receptors.

Where a transmitter is made matters. Acetylcholine, the amines, the amino acids and the purines are small molecules. They are made in the axon terminal and loaded into vesicles there. The enzymes that make them are proteins, so those are built in the cell body and shipped down the axon. Peptides are built on ribosomes like any other protein, so they are made in the cell body, packaged into vesicles there, and carried to the terminal by fast axonal transport along microtubules, which was covered in the lecture on neurons. Nitric oxide is made from the amino acid arginine by an enzyme. It lasts only a few seconds, and it acts on whichever cells it diffuses into.

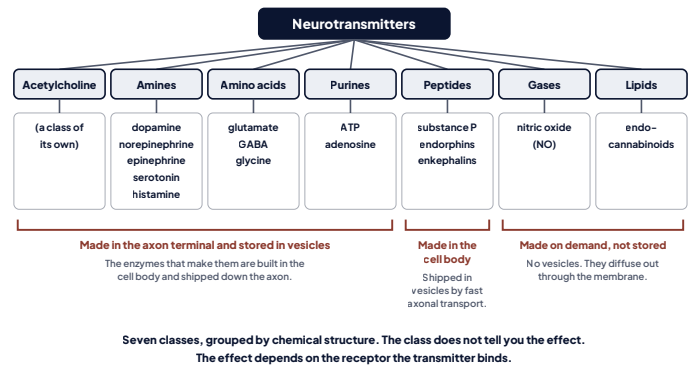


Figure 7. The seven classes, grouped by chemical structure, with where each one is made.

EACH TRANSMITTER, ITS USUAL EFFECT, AND ONE PLACE IT ACTS

TRANSMITTER	USUAL EFFECT	A REAL SITE OF ACTION
Acetylcholine	Excitatory at nicotinic receptors. Excitatory or inhibitory at muscarinic receptors.	Neuromuscular junction (excites skeletal muscle). Heart pacemaker (slows it). Autonomic ganglia and the brain.
Norepinephrine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Released by sympathetic neurons. Speeds the heart through beta-1 receptors and relaxes airway smooth muscle through beta-2 receptors. Arousal pathways in the brain.
Dopamine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Brain: pathways for movement in the basal nuclei, and for reward.
Serotonin	Either one. Most of its receptors are metabotropic.	Brainstem neurons that project widely through the brain: mood, sleep and appetite.
Glutamate	Excitatory. The main excitatory transmitter of the central nervous system.	Brain and spinal cord. Its AMPA and NMDA receptors are cation channels.
GABA	Inhibitory. The main inhibitory transmitter of the brain.	Brain. Its GABA-A receptor is a Cl^- channel, which holds the membrane away from threshold.
Glycine	Inhibitory. The main inhibitory transmitter of the spinal cord.	Spinal cord and brainstem. Its receptor is also a Cl^- channel.
Neuropeptides	Depends on the peptide. All of their receptors are metabotropic.	Substance P excites pain pathways in the spinal cord. Endorphins and enkephalins inhibit them.

WORKED PROBLEM 3

A motor neuron has its cell body in the spinal cord and its terminal in the foot, 1 m away. Fast axonal transport moves vesicles at up to about 400 mm per day. How long does a new supply of a peptide take to arrive? Compare that with the supply of acetylcholine.

Time is distance divided by speed. Keep the units the same.

THE WORK

- Convert the axon length to the units of the transport speed. 1 m = **1,000 mm**.
- Time = distance ÷ speed = 1,000 mm ÷ 400 mm per day = **2.5 days**. That is the fastest case. Any vesicle that moves more slowly takes longer.
- Now acetylcholine. The terminal makes it on the spot from choline and acetyl CoA, and about half of the choline is recovered from the cleft after each release. Emptied vesicle membrane is taken back in by endocytosis and refilled within about a minute.
- So the two supplies are refilled on very different time scales: about a minute for acetylcholine, and days for a peptide.
- Prediction: a neuron that fires rapidly for a long time can run low on its peptide long before it runs low on its small transmitter. A peptide used up at the terminal can only be replaced by the cell body, 2.5 days away.

Catecholamines

Dopamine, norepinephrine and epinephrine. All three are made from the amino acid tyrosine.

Glutamate

The main excitatory transmitter of the central nervous system.

GABA and glycine

The main inhibitory transmitters: GABA in the brain, glycine in the spinal cord.

Neuropeptide

A transmitter that is a short chain of amino acids, made in the cell body and shipped to the terminal.

Ending the signal: three routes of removal

SLIDES 13 TO 14 OF THE DECK

A signal carries information only if it ends. If transmitter stayed in the cleft, the receptors would stay bound, and the postsynaptic cell could not tell one action potential from the next. A doorbell whose button sticks tells you nothing about how many visitors there are. Three routes clear transmitter from the cleft, and each synapse relies mainly on one of them.

The first route is breakdown by an enzyme. Acetylcholinesterase is anchored in the cleft, and it splits acetylcholine into choline and acetate within milliseconds. Neither piece activates the receptor. A transporter carries the choline back into the terminal, where it is joined to acetyl CoA to make new acetylcholine.

The second route is reuptake. Transporter proteins in the presynaptic membrane carry whole transmitter molecules back into the terminal, where they are reloaded into vesicles or broken down. Serotonin, norepinephrine and dopamine are removed mainly this way. Glutamate and GABA are taken up by astrocytes as well as by neurons. These transporters are cotransporters that use the Na^+ gradient, the same kind of secondary active transport covered in Week 2.

The third route is diffusion. Some transmitter always drifts out of the open edges of the cleft. For peptides this is the main route, and peptidases outside the cleft then break them down.

Figure 9 shows what happens to the response to one action potential when removal is blocked, using the model from this lecture. Release is unchanged, so the same amount of transmitter enters the cleft. It then stays longer. Receptors that would have emptied are bound again and again, so the response keeps growing for longer, which makes it taller, and it takes longer to end, which makes it wider. In the model the peak goes from 1.00 to about 1.36 with reuptake blocked and to about 1.80 with the enzyme blocked. Those sizes are illustrative. Which block matters more at a real synapse depends on which route that synapse mainly uses.

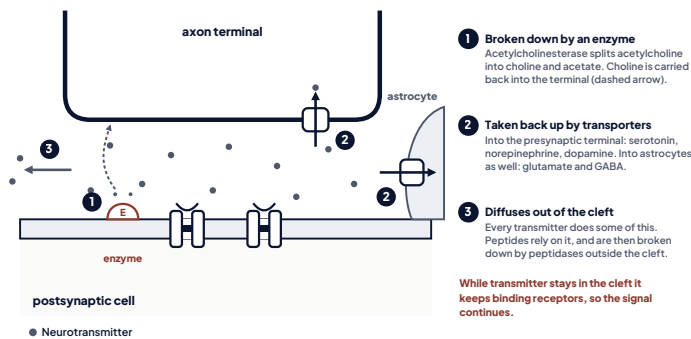


Figure 8. Breakdown by an enzyme, reuptake by transporters, and diffusion out of the cleft.

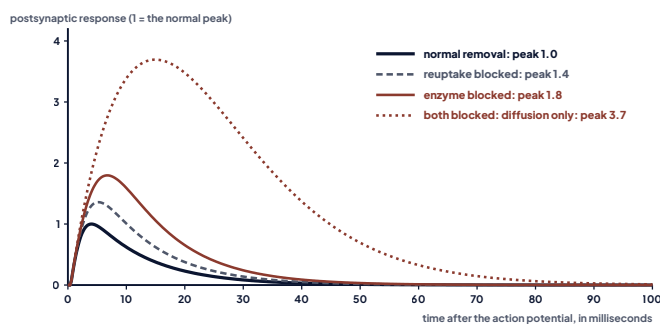


Figure 9. The response to one action potential with each removal route blocked. Relative units from the model. The shapes are illustrative.

Acetylcholinesterase

The enzyme in the cleft that splits acetylcholine into choline and acetate.

Reuptake

Transport of whole transmitter molecules back into the presynaptic terminal, or into glia.

Diffusion

Movement of transmitter out of the cleft down its concentration gradient.

Toxins and drugs as experiments on single steps SLIDES 16 TO 18 OF THE DECK

Much of what is known about synapses came from toxins and drugs that block one step at a time. For each one, find the step it blocks and then predict what the postsynaptic cell receives. Anything that blocks steps 1 to 8 gives the postsynaptic cell less signal. Anything that blocks step 9 gives it more, because removal is the only step whose job is to end the signal.

Curare and botulinum toxin both paralyze skeletal muscle, and they can be told apart by measuring acetylcholine release. With curare, acetylcholine is released in the normal amount, and it is the receptors that are occupied. With botulinum toxin no acetylcholine is released. Blocking acetylcholinesterase helps against curare, because curare competes with acetylcholine for the receptor and more acetylcholine displaces it. It cannot help against botulinum toxin, because there is no acetylcholine to preserve.

One honest note about SSRIs. Blocking the transporter raises serotonin in the cleft within hours, yet the improvement in mood usually takes several weeks. So the longer term changes that neurons make in response to the extra serotonin must matter too, and those are still being worked out. For this course, know the step that the drug blocks.

Worked problem 4 compares botulinum toxin with an organophosphate. They block opposite steps at the same synapses, and most of their signs are opposite. The exception is skeletal muscle, where both end in weakness.

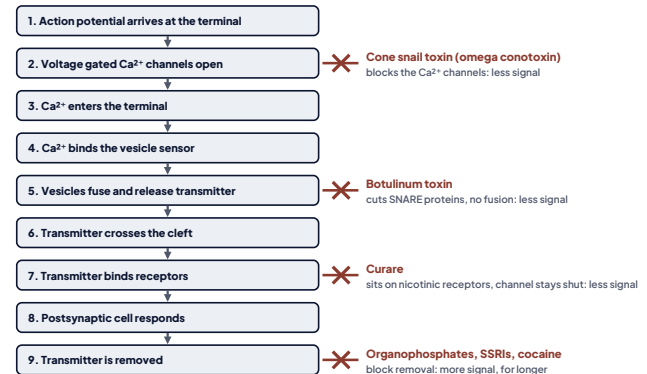


Figure 10. The nine steps from top to bottom, with the step each agent blocks.

SIX AGENTS, THE STEP EACH ONE BLOCKS, AND THE SIGNAL THE POSTSYNAPTIC CELL RECEIVES

AGENT	WHAT IT DOES	STEP	SIGNAL
Omega conotoxin, from cone snail venom	Blocks the voltage gated Ca^{2+} channels of the terminal	2	Less
Botulinum toxin	Cuts SNARE proteins, so vesicles cannot fuse	5	Less
Curare	Occupies nicotinic receptors without opening them	7	Less
Organophosphate insecticides and nerve agents	Inhibit acetylcholinesterase	9	More, for longer
SSRIs, such as fluoxetine and sertraline	Block the serotonin reuptake transporter	9	More, for longer
Cocaine	Blocks the dopamine reuptake transporter	9	More, for longer

WORKED PROBLEM 4

Botulinum toxin blocks release. An organophosphate blocks removal. Both act at every synapse that uses acetylcholine. Explain the signs of each at skeletal muscle, where the receptors are nicotinic, and at glands, gut, pupils and heart, where they are muscarinic, and explain why both end in muscle weakness.

Work out the acetylcholine in the cleft first. Then ask what each kind of target does with no signal, and with a signal that never ends.

THE WORK

- 1** **Botulinum toxin** acts at step 5. No vesicles fuse, so the acetylcholine in the cleft falls toward zero at every synapse that releases acetylcholine.
- 2** Skeletal muscle (nicotinic receptors) receives no signal: the muscles are weak and floppy. This is called flaccid paralysis. Glands and gut (muscarinic receptors) receive no signal either: dry mouth, dry eyes, constipation, and pupils that are wide and slow to react.
- 3** **An organophosphate** acts at step 9. Acetylcholinesterase is inhibited, so acetylcholine stays in the cleft and keeps binding at every one of the same synapses.
- 4** Glands, gut, airways and heart (muscarinic receptors) are stimulated continuously: saliva, tears, sweat and airway secretions increase, the gut is overactive, the airways narrow, the pupils become very small, and the heart rate slows.
- 5** Skeletal muscle (nicotinic receptors) is stimulated continuously too. At first the muscles twitch. Then they become weak. The reason is in the action potential lecture: a membrane that is held depolarized keeps its voltage gated sodium channels inactivated, so the muscle fiber can no longer fire.
- 6** So the two poisons give opposite signs at glands, gut, pupils and heart, and yet both can stop breathing, because a skeletal muscle needs a signal that turns on and then turns off.

The patient, explained

SLIDE 19 OF THE DECK

The baby has infant botulism. Honey can carry spores of the bacterium *Clostridium botulinum*. An adult's gut bacteria keep the spores from growing. An infant's gut bacteria are not yet established, so the spores grew in her large intestine and the bacteria made botulinum toxin. Honey is one known source of the spores. Soil and dust are another, and in many cases the source is never found.

The toxin was absorbed into her blood and taken up by axon terminals that release acetylcholine. It does not reach the brain in meaningful amounts, which is why she stayed alert and could feel. Inside the terminal, the toxin is an enzyme that cuts SNARE

proteins. Action potentials still arrived, Ca^{2+} channels still opened and Ca^{2+} still entered. The vesicles could not fuse, so no acetylcholine was released. Step 5 failed.

Every synapse that releases acetylcholine went quiet. The gut slowed first, which was the constipation. Then skeletal muscles stopped receiving a signal: the weak suck and cry, the drooping eyelids, the floppy neck, and last the diaphragm.

Treat it as an experiment. The variable that failed is the acetylcholine released per action potential. An antibody against the toxin binds toxin that is still in the blood, so it cannot enter any more terminals. It cannot reach toxin that is already inside terminals, so it stops the illness from progressing and does not reverse the weakness that is already there. A ventilator does the work of the diaphragm in the meantime. Recovery depends on the neurons themselves: they make new SNARE proteins and grow new terminal branches, which takes weeks to months. Infants who are supported through that time usually recover fully.

Compare this with organophosphate poisoning, where the opposite step fails. There, removal is blocked, acetylcholine stays in the cleft, and the signs at glands, gut, pupils and heart are the opposite ones.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Proteins change shape when something binds to them. That is how a receptor or a channel opens.

An enzyme speeds up one reaction and is not used up by it.

Diffusion moves molecules from higher to lower concentration, and it is fast only over very short distances.

Chemistry review page

FROM EARLIER LECTURES

Resting potential lecture: E_{Ca} is about +123 mV, free Ca^{2+} is about 1 mM outside and 0.0001 mM inside, and ligand gated channels open when a molecule binds.

Neurons and glia lecture: the axon terminal is the output zone, fast axonal transport carries vesicles from the cell body, and astrocytes take up transmitter.

Action potential lectures: voltage gated channels, inactivation of sodium channels, graded potentials, and the frequency code.

Week 3: exocytosis, endocytosis and secondary active transport.

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

The Nernst equation with $z = +2$, which needs a calculator with a log button.

Time = distance ÷ speed, keeping the units the same.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

21

Sequence at a chemical synapse

Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

Do one of the two. Head your paper with the number.

21-1

Draw one axon terminal big enough to fill half your space, with the cleft and a patch of postsynaptic membrane. Number the events in order from the action potential arriving to the postsynaptic channel opening, one arrow per step, and show where calcium enters and what it does once inside. Then redraw the terminal sitting in a calcium free bath. One line: name the step where the chain stops, and say what still happens normally upstream of it.

21-2

Draw two traces stacked on one shared time axis in milliseconds, the presynaptic action potential on top and the postsynaptic membrane potential below. Mark the synaptic delay between them, about 0.5 ms. In the gap under that delay, draw the events that are happening while nothing shows on either trace. One line: which single step you think eats most of that delay, and why.

22**Neurotransmitter classes**

Group neurocrines into their seven chemical classes, acetylcholine, amines, amino acids, peptides, purines, gases, and lipids, and match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

Do one of the two. Head your paper with the number.

22-1

Draw a rough brain, a spinal cord, and one peripheral effector, side by side. Place acetylcholine, a catecholamine, serotonin, glutamate, GABA, glycine, and a neuropeptide at a real site of action on your drawing. Flag each one excitatory or inhibitory with an up or down arrow drawn on the postsynaptic membrane. One line: name the one you had to flag both ways, and say what forced that.

22-2

Draw acetylcholine arriving at two postsynaptic membranes side by side, a skeletal muscle fiber with nicotinic receptors and a cardiac pacemaker cell with muscarinic receptors. Show the ion moving at each and draw the membrane potential trace under each. Same transmitter, opposite result. One line: given both drawings, state what actually decides whether a synapse is excitatory or inhibitory.

23**Neurotransmitter removal**

Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

Do one of the two. Head your paper with the number.

23-1

Draw a synaptic cleft with transmitter still in it, then draw all three exit routes off that cleft, enzymatic breakdown, reuptake, and simple diffusion. Show where each one physically happens, on the postsynaptic side, on the presynaptic side, or out the edges. Name the transmitter you picked and put a real drug on the route it blocks. One line: what the postsynaptic cell experiences once that route is shut.

23-2

Draw the postsynaptic potential from one normal release event, membrane potential on the y axis and time on the x. On the same axes draw that trace again under two drugs, one that blocks degradation and one that blocks reuptake. Show what changes, the height, the width, or both. One line: which drug you expect to change the trace more, and what you would need to measure to be sure.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The chemical synapse. Concentrations and voltages are typical teaching values. All figures are original drawings for this course.